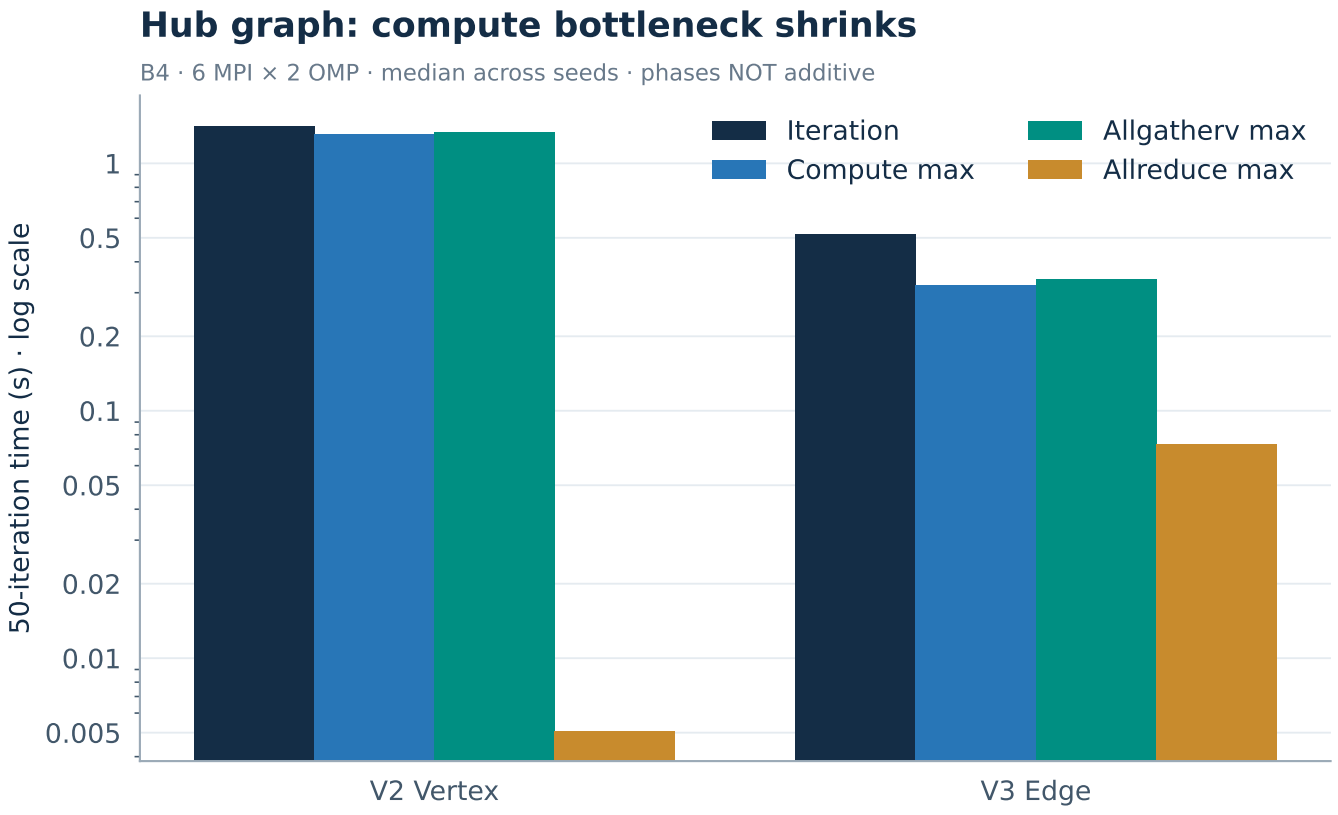
Hub graph: compute bottleneck shrinks
B4 · 6 MPI × 2 OMP · median across seeds · phases NOT additive
Iteration
Compute max
Allgatherv max
Allreduce max
50-iteration time (s) · log scale
1
0.5
0.2
0.1
0.05
0.02
0.01
0.005
V2 Vertex
V3 Edge